

Important Links

Cross-Generator Generalization in Deepfake Detection | IE7374: Generative AI

Dataset access: FaceForensics++

- **Request form (start here):** https://docs.google.com/forms/d/e/1FAIpQLSdRRR3L5zAv6tQ_CKxmK4W96tAab_pfBu2EKAgQbeDVhmXagg/viewform
- **Official repository:** <https://github.com/ondyari/FaceForensics>
- **Download and README (script, sizes):**
<https://github.com/ondyari/FaceForensics/blob/master/dataset/README.md>
- **Terms of Use:** http://kaldir.vc.in.tum.de/faceforensics_tos.pdf
- **Contact:** faceforensics@googlegroups.com
- **Kaggle mirror (unofficial):** <https://www.kaggle.com/datasets/xdxd003/ff-c23>
- **Kaggle mirror (unofficial):** <https://www.kaggle.com/datasets/hungle3401/faceforensics>

Kaggle mirrors are community re-uploads. Confirm they include all four methods at c23 before relying on them; the official form is the license-clean route.

Dataset access: DFDC (optional stretch)

- **Meta DFDC page:** <https://ai.meta.com/datasets/dfdc/>
- **Kaggle competition (download):** <https://www.kaggle.com/c/deepfake-detection-challenge>
- **DFDC Preview paper (about 5,000 clips):** <https://arxiv.org/abs/1910.08854>
- **DFDC full paper:** <https://arxiv.org/abs/2006.07397>

Core papers

- **FaceForensics++ (Rossler et al., 2019):** <https://arxiv.org/abs/1901.08971>
- **XceptionNet (Chollet, 2017):** <https://arxiv.org/abs/1610.02357>
- **EfficientNet (Tan and Le, 2019):** <https://arxiv.org/abs/1905.11946>
- **LipForensics, cross-manipulation numbers (Haliassos et al., 2021):** <https://arxiv.org/abs/2012.07657>
- **Celeb-DF, cross-dataset collapse (Li et al., 2020):** <https://arxiv.org/abs/1909.12962>
- **DeepfakeBench, reproducible AUCs (Yan et al., 2023):** <https://arxiv.org/abs/2307.01426>

Supporting papers (background)

- **Face X-Ray (Li et al., 2020):** <https://arxiv.org/abs/1912.13458>
- **MesoNet (Afchar et al., 2018):** <https://arxiv.org/abs/1809.00888>
- **Frequency artifacts (Frank et al., 2020):** <https://arxiv.org/abs/2003.08685>
- **Diffusion deepfake detection (Ricker et al., 2024):** <https://arxiv.org/abs/2210.14571>

Code and models

- **DFDC winning solution (EfficientNet-B7 + MTCNN):** https://github.com/selimsef/dfdc_deepfake_challenge
- **DeepfakeBench repository (standard pipeline):** <https://github.com/SCLBD/DeepfakeBench>
- **timm, EfficientNet and Xception pretrained weights:** <https://github.com/huggingface/pytorch-image-models>
- **PyTorch cu128 wheels (RTX 5070 and P100):** <https://download.pytorch.org/whl/cu128>

Verify the DeepfakeBench organization name if the link does not resolve; the arXiv paper above is the verified reference.

Full citations and per-fact sources are in `.planning/research/` (BACKGROUND.md, DATASETS.md, METHODOLOGY_CNN.md).